

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

The minimum energy required by a hydrogen atom in ground state to emit radiation in Balmer series is nearly:

- (1) 1.5eV
- (2) 13.6eV
- (3) 1.9eV
- (4) 12.1eV

Q2 - 2024 (01 Feb Shift 2)

A particular hydrogen - like ion emits the radiation of frequency 3×10^{15} Hz when it makes transition from $n = 2$ to $n = 1$. The frequency of radiation emitted in transition from $n = 3$ to $n = 1$ is $\frac{x}{9} \times 10^{15}$ Hz, when $x =$ _____

Q3 - 2024 (27 Jan Shift 1)

The radius of third stationary orbit of electron for Bohr's atom is R. The radius of fourth stationary orbit will be:

- (1) $\frac{4}{3}R$
- (2) $\frac{16}{9}R$
- (3) $\frac{3}{4}R$
- (4) $\frac{9}{16}R$

Q4 - 2024 (27 Jan Shift 2)

If Rydberg's constant is R, the longest wavelength of radiation in Paschen series will be $\frac{\alpha}{7R}$, where $\alpha =$ _____

Q5 - 2024 (29 Jan Shift 1)

When a hydrogen atom going from $n = 2$ to $n = 1$ emits a photon, its recoil speed is $\frac{x}{5}$ m/s. Where $x =$ _____. (Use : mass of hydrogen atom $= 1.6 \times 10^{-27}$ kg)

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Q6 - 2024 (29 Jan Shift 2)

Given below are two statements :

Statement I : Most of the mass of the atom and all its positive charge are concentrated in a tiny nucleus and the electrons revolve around it, is Rutherford's model.

Statement II : An atom is a spherical cloud of positive charges with electrons embedded in it, is a special case of Rutherford's model.

In the light of the above statements, choose the most appropriate from the options given below.

(1) Both statement I and statement II are false

(2) Statement I is false but statement II is true

(3) Statement I is true but statement II is false

(4) Both statement I and statement II are true

Q7 - 2024 (29 Jan Shift 2)

Hydrogen atom is bombarded with electrons accelerated through a potential different of V , which causes excitation of hydrogen atoms. If the experiment is being formed at $T = 0$ K. The minimum potential difference needed to observe any Balmer series lines in the emission spectra will be $\frac{\alpha}{10} V$, where $\alpha =$ _____

Q8 - 2024 (30 Jan Shift 1)

The ratio of the magnitude of the kinetic energy to the potential energy of an electron in the 5th excited state of a hydrogen atom is :

(1) 4

(2) $\frac{1}{4}$

(3) $\frac{1}{2}$

(4) 1

Q9 - 2024 (30 Jan Shift 1)

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A electron of hydrogen atom on an excited state is having energy $E_n = -0.85\text{eV}$. The maximum number of allowed transitions to lower energy level is

Q10 - 2024 (30 Jan Shift 2)

An electron revolving in n^{th} Bohr orbit has magnetic moment μ_n . If $\mu_n \propto n^x$, the value of x is:

- (1) 2
- (2) 1
- (3) 3
- (4) 0

Q11 - 2024 (31 Jan Shift 1)

The mass defect in a particular reaction is 0.4 g. The amount of energy liberated is $n \times 10^7 \text{kWh}$, where $n =$

(speed of light = $3 \times 10^8 \text{ m/s}$)

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Questions with Answer Keys

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Answer Key

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 Q1 (4)	 mathongo	 Q2 (32)	 Q3 (2)	 Q4 (144)	 mathongo
 Q5 (17)	 mathongo	 Q6 (3)	 Q7 (121)	 Q8 (3)	 mathongo
 Q9 (6)	 mathongo	 Q10 (2)	 Q11 (1)	 mathongo	 mathongo
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Solutions

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Q1

Transition from $n = 1$ to $n = 3$ $\Delta E = 12.1 \text{ eV}$

Q2

$$E = -13.6z^2 \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right)$$

$$E = C \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

$$hv = C \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

$$\frac{v_1}{v_2} = \frac{\left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]_{2-1}}{\left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]_{3-1}}$$

$$= \frac{\left[\frac{1}{1} - \frac{1}{4} \right]}{\left[\frac{1}{1} - \frac{1}{9} \right]} = \frac{3/4}{8/9}$$

$$= \frac{3}{4} \times \frac{9}{8}$$

$$\frac{v_1}{v_2} = \frac{27}{32}$$

$$v_2 = \frac{32}{27} v_1 = \frac{32}{27} \times 3 \times 10^{15} \text{ Hz} = \frac{32}{9} \times 10^{15} \text{ Hz}$$

Q3

$$r \propto \frac{n^2}{Z}$$

$$\frac{r_4}{r_3} = \frac{4^2}{3^2}$$

$$r_4 = \frac{16}{9} R$$

Q4

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Solutions

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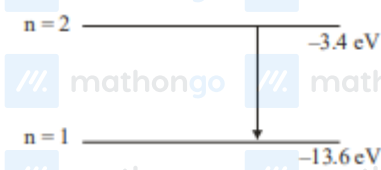
Longest wavelength corresponds to transition between $n = 3$ and $n = 4$

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{3^2} - \frac{1}{4^2} \right) = RZ^2 \left(\frac{1}{9} - \frac{1}{16} \right)$$

$$= \frac{7RZ^2}{9 \times 16}$$

$$\Rightarrow \lambda = \frac{144}{7R} \text{ for } Z = 1 \quad \therefore \alpha = 144$$

Q5



$$\Delta E = 10.2 \text{ eV}$$

$$\text{Recoil speed (v)} = \frac{\Delta E}{mc}$$

$$= \frac{10.2 \text{ eV}}{1.6 \times 10^{-27} \times 3 \times 10^8}$$

$$= \frac{10.2 \times 1.6 \times 10^{-19}}{1.6 \times 10^{-27} \times 3 \times 10^8}$$

$$v = 3.4 \text{ m/s} = \frac{17}{5} \text{ m/s}$$

$$\text{Therefore, } x = 17$$

Q6

According to Rutherford atomic model, most of mass of atom and all its positive charge is concentrated in tiny nucleus & electron revolve around it.

According to Thomson atomic model, atom is spherical cloud of positive charge with electron embedded in it.

Hence,

Statement I is true but statement II false.

Q7

For minimum potential difference electron has to make transition from $n = 3$ to $n = 2$ state but first electron has to reach to $n = 3$ state from ground state. So, energy of bombarding electron should be equal to energy difference of $n = 3$ and $n = 1$ state.

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Solutions

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$$\Delta E = 13.6 \left[1 - \frac{1}{3^2} \right] e = eV$$

$$\frac{13.6 \times 8}{9} = V$$

$$V = 12.09 \text{ V} \approx 12.1 \text{ V}$$

$$\text{So, } \alpha = 121$$

Q8

$$\frac{1}{2}|PE| = KE \text{ for each value of } n \text{ (orbit)}$$

$$\therefore \frac{KE}{|PE|} = \frac{1}{2}$$

Q9

$$E_n = -\frac{13.6}{n^2} = -0.85$$

$$\Rightarrow n = 4$$

No of transition

$$= \frac{n(n-1)}{2} = \frac{4(4-1)}{2} = 6$$

Q10

$$\text{Magnetic moment} = i\pi r^2$$

$$\mu = \frac{evr}{2}$$

$$\mu \alpha \left(\frac{1}{n} \right) n^2$$

$$\frac{\mu \alpha n}{x} = 1$$

Q11

$$E = \Delta mc^2$$

$$= 0.4 \times 10^{-3} \times (3 \times 10^8)^2$$

$$= 3600 \times 10^7 \text{ kWs}$$

$$= \frac{3600 \times 10^7}{3600} \text{ kWh} = 1 \times 10^7 \text{ kWh}$$

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